R1.4 Entropy and spontaneity (HL)

Entropy (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Predict whether a physical or chemical change will result in an increase or decrease in entropy of a system.
- Calculate standard entropy changes, $\Delta S^{\ominus}$, from standard entropy values, $S^{\ominus}$.

In our daily lives, we encounter many other instances of increased disorder. Think of an ice cube melting: the particles that were once organised in the cube, are now dispersing as a liquid. Perhaps you have heard about statues made of calcium carbonate, $CaCO_3$, that are slowly degrading due to exposure to acid rain.

Credit: Shenki, Getty Images (https://www.gettyimages.com/detail/photo/ice-cube-royalty-free-image/104789662)

Credit: Photography by Jason Gallant, Getty Images (https://www.gettyimages.com/detail/photo/statue-at-angkor-thom-north-gate-royalty-free-image/969274980)

Figure 1. Water disperses when ice melts and calcium carbonate statues dissolve when exposed to acid rain.

Have you noticed that in all the examples described previously, the tendency is for an increase in disorder? Why is that?

Entropy

In all these examples there is an increase in entropy.

Entropy (*S*) can be thought of as a measure of disorder or randomness in a system. It specifically refers to the distribution of energy among the particles in a system. Consequently, this means that in a system with higher entropy, the particles are more spread out and moving more. Imagine a closed jar where you placed coloured sand in beautifully organised layers. The smallest movement of the jar will cause the grains of sand to move and mix. The more the jar is moved, the more the colours will mix and they will never return to their original arrangement, no matter how many times you shake the jar. The system tends to remain in randomness as this means the energy is more dispersed.

Recall from section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) when you learned about solids, liquids and gases and how the particles were arranged. Which state of matter seems the most ordered to you?

In **Interactive 1** below, order the three systems according to an increasing level of entropy (or disorder).

Increasing entropy →

Check

Interactive 1. Increasing entropy.

Nature of Science

Aspect: Patterns and trends

The laws of thermodynamics have changed in content and numbering through history, combining the ideas of chemists and physicists to make sense of temperature, energy and entropy.

The first and second laws of thermodynamics were developed by various scientists in the 19th century. The third law is more recent and was formulated by a physical chemist called Walter Nernst in 1906. It states that as the temperature of a pure substance reaches zero kelvin (absolute zero), its entropy will also reach a minimum. The increasing understanding of the way matter behaves at extremely low temperatures continues to lead to various scientific discoveries in the field of low-temperature physics, such as the discovery of superconductivity and superfluidity.

Interactive 2 shows the relative entropy of solids, liquids, and gases. Under the same conditions, solids (with their fixed arrangements of particles) have the lowest entropy, and gases (with their random arrangements of particles) have the highest entropy. In gases, energy can be distributed in more ways than in liquids and solids – they have higher entropy.

Interactive 2. Entropy and the States of Matter.

More information for interactive 2

An interactive slide illustrates the relative entropy of solids, liquids, and gases.
The structure of the states of matter is as follows:
Solid: Particles are tightly packed in a fixed, orderly arrangement.
Liquid: Particles are close together but not in a fixed arrangement.
Gases: Particles are loosely packed and spread out randomly.
Entropy increases from solid to liquid and then to gas, as indicated by a rightward arrow at the top.
Entropy decreases from gas to liquid and then to solid, as indicated by a leftward arrow at the bottom.
This interactive slide helps users understand the relationship between the states of matter and entropy and the particle arrangement in solids, liquids, and gases.

International Mindedness

One of the major issues that we face on the planet currently is sustainability. Various countries around the world found the need to create protected areas both on land and on the ocean to preserve and protect ecosystems. Marine protected areas, for example, are designated areas in the ocean where human activities are regulated, promoting sustainable fisheries, protecting vulnerable marine areas, and conserving biodiversity.

In order to have sustainable systems the focus is on maintaining their order and structure over time. You might now be pausing to consider how your understanding of the concept of entropy can impact your perspective on this issue.

As you learned, all systems tend to increase in entropy, which is the tendency for systems to become disordered and unstable. In order to promote sustainability worldwide, it is required an understanding of the importance of sustainability and the need for global cooperation to address environmental issues and promote sustainable development. The study of entropy can help us appreciate the challenges involved in creating sustainable global systems and the need for coordinated efforts to address these challenges.

Changes in entropy (ΔS) for physical changes

Recall from section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) that the state of a substance depends on its temperature. Solid states exist at lower temperatures and gas states at higher temperatures. If the temperature of a substance changes, its state might also change.

During changes of state, there is a change in the entropy of a system. When an ice cube melts, the water molecules that were compacted neatly in a crystalline shape start moving past each other. This represents an increase in entropy as the kinetic energy of the particles increases.

Video 1. An Ice Cube Melting Is Undergoing a Change in Entropy.

Credit: dogholiday, Getty Images (https://www.gettyimages.com/detail/video/melting-ice-cube-time-lapse-stock-footage/1064566246)

More information for video 1

An interactive video demonstrates the melting process of an ice cube placed on a wooden surface. The ice cube is transparent but contains a few air bubbles trapped inside.

As the ice gradually melts, it transforms into liquid water. The water spreads across the wooden surface of the wood, seeping into the natural cracks of the wood.

A certain amount of water content on the outside slowly evaporates, leaving behind moisture traces.

This interactive helps users understand the physical changes of water, including melting and evaporation.

The play button with a slider is provided at the bottom of the video for the user to play and pause the video as required. The speedometer, next to the slider, with playback rates 0.25, 0.5, 1, 1.25, 1.5, and 2 allows the user to adjust the speed of the video. The sound icon, next to this, allows users to play the video with sound or in mute. The zoom-out at the bottom right enables the video to be played in full-screen mode

In the reverse process, when water freezes the particles that were able to move past each other in the liquid water get trapped into a crystalline structure and ice forms. In this case, there was an increase in order, which means a decrease in entropy.

Figure 2 shows a summary of the changes of state and the corresponding changes of entropy.

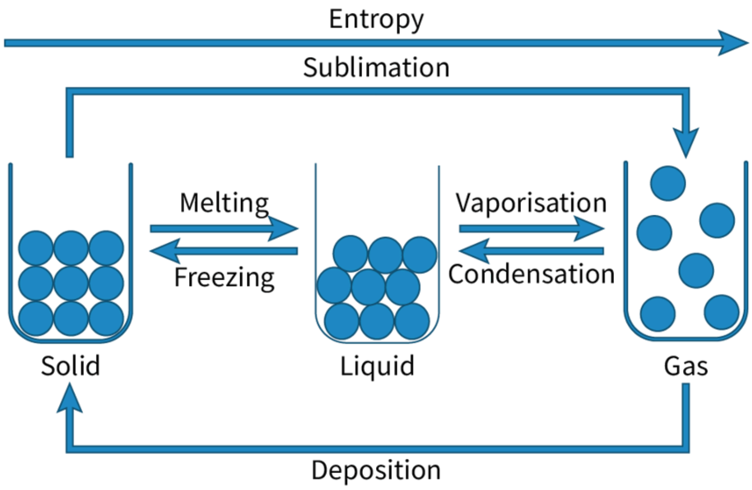


Figure 2. Changes of state and corresponding entropy.

More information for figure 2

The diagram illustrates the changes in states of matter and the corresponding changes in entropy. It highlights three main states: solid, liquid, and gas, each represented by circles in different containers.

- **Solid to Liquid**: The diagram shows an arrow labeled 'Melting' pointing from solid to liquid and an arrow labeled 'Freezing' in the opposite direction, indicating the reversible process between these states.
- **Liquid to Gas**: Similarly, arrows labeled 'Vaporization' and 'Condensation' illustrate the reversible transition between liquid and gas states.
- **Solid to Gas**: The process of 'Sublimation' is represented by an arrow pointing directly from solid to gas.
- **Gas to Solid**: The arrow labeled 'Deposition' shows the direct transition of gas to solid.

At the top of the diagram, there is an overarching arrow labeled 'Entropy,' indicating that transition processes affect entropy levels, with gas assumed to have the highest entropy compared to solid and liquid based on common scientific understanding.

[Generated by AI]

Entropy change (ΔS) can be negative ($-\Delta S$) or positive ($+\Delta S$), depending on whether there is a decrease in entropy or an increase in entropy. When considering the sign of the entropy change (positive or negative), it is important to consider the changes in the state of the reactants and products. For example, evaporation results in an increase in entropy ($+\Delta S$) as gases have higher entropy than liquids. In chemical reactions that involve solids or liquids being converted into gaseous products, there will be a large increase in entropy.

$$H_2O(l) + \text{heat} \rightarrow H_2O(g)$$

Recall the example with layers of coloured sand in a jar. Think of each colour of sand as a pure substance. After the container is shaken, you have made a mixture. The combination of pure substances to form homogeneous or heterogeneous mixtures, also leads to an increase in entropy. For example, when you place sodium chloride (solute) into water (solvent), the solid salt dissolves, which means its ions become dispersed between the water molecules – you have created a solution. This represents an increase in the entropy of the system, $+\Delta S$, as shown in **Figure 3**.

$$NaCl(s) \rightarrow Na^{+}(aq) + Cl^{-}(aq)$$

This process can also be represented by the equation:

$$NaCl(s) \rightarrow NaCl(aq)$$

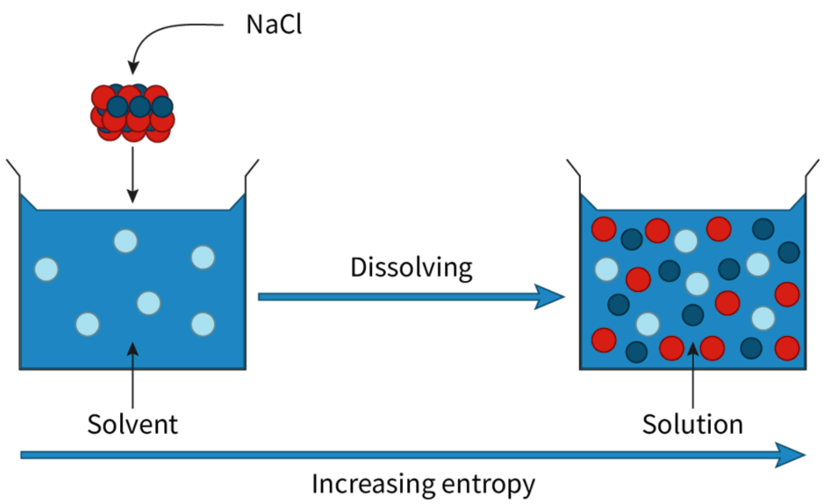


Figure 3. Entropy of a dissolution.

More information for figure 3

The image is a diagram illustrating the process of dissolving NaCl (sodium chloride) in water and the resulting increase in entropy. On the left, there's a container labeled 'Solvent' containing water with a few light blue circles representing water molecules. Above, there is a cluster of red and dark blue circles labeled 'NaCl' representing the solid salt, with an arrow indicating it is being introduced into the water. In the center, the word 'Dissolving' is written with an arrow pointing to the right.

To the right, a second container labeled 'Solution' contains a mix of red, light blue, and dark blue circles, indicating that the salt has dissolved in the water, creating a homogenous solution. Below the diagram, a horizontal arrow labeled 'Increasing entropy' signifies the increase in entropy during the dissolution process.

[Generated by AI]

For the reverse process, if the solvent is evaporated and the salt crystallised there is a decrease in the entropy of the system, $-\Delta S$.

Use the interactive to classify the changes of entropy (increase/decrease) in the examples of processes that result in a change in entropy.

Worked example 1

Examine the equation below that shows the condensation of water.

$$H_2O(g) \rightarrow H_2O(l)$$

Predict if there is an increase or decrease in entropy and whether the sign for the change in entropy is positive or negative.

As there is a change from gas to liquid, there is a decrease in entropy. ΔS is therefore negative.

Worked example 2

Examine the equation for dissolving sodium hydrogen carbonate to form a solution.

$$NaHCO_3(s) \rightarrow NaHCO_3(aq)$$

Predict if there is an increase or decrease in entropy and the sign for the change in entropy.

As there is a change from solid to a solution. There is an increase in entropy, so ΔS is positive.

Worked example 3

Write the equation, including state symbols, for the deposition of carbon dioxide to create dry ice.

Predict if there is an increase or decrease in entropy and the sign for the change in entropy.

$$CO_2(g) \rightarrow CO_2(s)$$

As there is a change from gas to solid, there is a decrease in entropy. ΔS is therefore negative.

Worked example 4

Write the equation, including state symbols, for the vaporisation of ethanol, C_2H_5OH.

Predict if there is an increase or decrease in entropy and the sign for the change in entropy.

$$C_2H_5OH(l) \rightarrow C_2H_5OH(g)$$

As there is a change from liquid to gas, there is an increase in entropy. ΔS is therefore positive.

Creativity, activity, service

Strand: Service
Learning outcomes:

- Demonstrate engagement with issues of global significance.
- Demonstrate how to initiate and plan a CAS experience.

The melting of ice is an example of a physical process that results in an increase in entropy. Can you think of a global environmental issue that involves the melting of ice?

According to Climate.gov (https://www.climate.gov/news-features/understanding-climate/climate-change-global-sea-level#:~:text=Global%20mean%20sea%20level%20has,of%20seawater%20as%20it%20warms. the global sea level has risen by approximately 20 cm since 1880. This is caused by the melting of the polar ice caps. For example, according to NASA (https://climate.nasa.gov/vital-signs/ice-sheets/#:~:text=Antarctica%20is%20losing%20ice%20mass,adding%20to%20sea%20level%2C Antarctica is losing 1.4×10^{14} kg of ice each year.

A possible CAS experience is to research the causes and effects of the changing sea levels and raise awareness of this issue in your school or the local community.

Changes in entropy (ΔS) in chemical reactions

In the examples above, only physical changes were considered. However, there can also be a change in entropy in chemical reactions. An example is the decomposition of hydrogen peroxide:

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

From the balanced equation, we can see that two moles of solution in the reactants become two moles of liquid plus one mole of gas in the products, which represents an increase in entropy. A change in the amount (in mol) of gaseous reactants or products can be used to predict a change in entropy. An increase in the number of moles of gas creates an increase in entropy and a decrease in the number of moles of gas on the products side signifies a decrease in entropy.

It is also important to note that if there are the same number of moles of gas in the reactants and products, then the entropy change will be quite small. Here is an example of a synthesis reaction of hydrochloric acid:

$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$$

From the balanced equation, we can see that two moles of gas in the reactants become two moles of gas in the products, which represents a minimal and unknown change in entropy.

Worked example 5

Examine the equation below for the decomposition of calcium carbonate.

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

Predict if there is an increase or decrease in entropy and state the sign for the change in entropy.

There is one mole of gas in the products and only solids in the reactants. There is an increase in entropy; ΔS is therefore positive.

Worked example 6

Examine the equation below for the synthesis of magnesium oxide.

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

Predict if there is an increase or decrease in entropy and state the sign for the change in entropy.

There is one mole of gas in the reactants and only solids in the product. As there is a change from gas to solid, there is a decrease in entropy. ΔS is therefore negative.

Worked example 7

Write the equation, including state symbols, for the formation of water from oxygen gas and hydrogen gas.

Predict if there is an increase or decrease in entropy and state the sign for the change in entropy.

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

There are three moles of gas in the reactants and only liquid in the product. As there is a change from gas to liquid, there is a decrease in entropy. ΔS is therefore negative.

Worked example 8

Write the equation, including state symbols, for the decomposition of ammonium nitrate powder, NH_4NO_3, into nitrous oxide gas (N_2O) and water vapour.

Predict if there is an increase or decrease in entropy and state the sign for the change in entropy.

$$NH_4NO_3(s) \rightarrow N_2O(g) + 2H_2O(g)$$

There are three moles of gas in the products and only solids in the reactants. As there is a change from solid to gas, there is an increase in entropy. ΔS is therefore positive.

Standard entropy values ($S^\ominus$) and changes in entropy ($\Delta S^\ominus$)

The standard entropy of a substance is the entropy change from heating the substance from absolute zero (0 K) to the standard temperature of 298 K. A perfect crystal at absolute zero (0 K) has a standard entropy of zero. Therefore, all substances have a positive standard entropy value.

Making connections

Subtopic S1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-40864/)—Why is the entropy of a perfect crystal at 0 K predicted to be zero?

As you learned in subtopic S1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-40864/), the lower the temperature of a substance, the more its particles tend to occupy a fixed position in a lattice structure, where they vibrate in place. The more order exists in that crystalline structure, the lower the entropy, as you have been learning in this section.

At absolute zero (0 K), all atomic motion stops, and the system is in a state of perfect order, with all atoms occupying specific positions in a regular lattice structure. Entropy is a measure of the number of possible arrangements or configurations of the system's particles. At absolute zero, the system is in its lowest energy state, and there is only one possible arrangement of particles that corresponds to that energy state. Therefore, the number of possible arrangements is zero, and the entropy is also zero.

In practice, it is not possible to cool a crystal to absolute zero (−273.15 °C), but it has been confirmed experimentally that as the temperature approaches zero kelvin, the entropy approaches zero. In October 2021, scientists broke the record for the coldest temperature ever measured in a lab. They achieved the bone-chilling temperature of 38 trillionths of a degree above −273.15 °C by dropping magnetised gas 120 metres (393 feet) down a tower.

Standard entropy values can be used to calculate the standard entropy change ($\Delta S^{\ominus}$) for a reaction, using the following equation:

$$\Delta S^{\ominus} = \Sigma S^{\ominus}\text{ (products)} - \Sigma S^{\ominus}\text{ (reactants)}$$

Study skills

Standard entropy values ($S^{\ominus}$) can be found in the DP chemistry data booklet in section 13 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/). Note that its units are J K^{-1} mol^{-1}.

Worked example 9

Consider the dissolution of sodium chloride in water.

$$NaCl(s) \rightarrow NaCl(aq)$$

Standard entropy values:

$S^{\ominus}$ (NaCl(s)) = +72.1 J K^{-1} mol^{-1}

$S^{\ominus}$ (NaCl(aq)) = +115.5 J K^{-1} mol^{-1}

Using the values above, calculate the standard entropy change for the reaction.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$S^{\ominus}\text{ (NaCl(s))} = +72.1\text{ J K}^{-1}\text{ m}$ $S^{\ominus}\text{ (NaCl(aq))} = +115.5\text{ J K}^{-}$
Step 2: Write out the equation.	$\Delta S^{\ominus} = \Sigma S^{\ominus}\text{ (products)} - \Sigma S$
Step 3: Substitute the values given.	$= 115.5 - 72.1$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding	$= +43.4\text{ J K}^{-1}\text{ mol}^{-1}$ Note that the positive sign show solute particles separate and dis

Worked example 10

Consider the physical change of bromine from liquid to gas.

$$Br_2(l) \rightarrow Br_2(g)$$

Standard entropy values:

$S^\ominus$ (Br_2(l)) = +152 J K^{-1} mol^{-1}

$S^\ominus$ (Br_2(g)) = +245.5 J K^{-1} mol^{-1}

Using the values above, calculate the standard entropy change for the reaction.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$S^\ominus\ (\mathrm{Br_2(l)}) = +152\ \mathrm{J\ K^{-1}\ mol^{-}}$ $S^\ominus\ (\mathrm{Br_2(g)}) = +245.5\ \mathrm{J\ K^{-1}\ m}$
Step 2: Write out the equation.	$\Delta S^\ominus = \Sigma S^\ominus\ (\mathrm{products}) - \Sigma S$
Step 3: Substitute the values given.	$= 245.5 - 152$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$= +93.5\ \mathrm{J\ K^{-1}\ mol^{-1}}$ Note that the positive sign show bromine moves from the liquid p

Worked example 11

Consider the reaction of but-1-ene (C_4H_8) with steam to form butanol (C_4H_9OH):

$$C_4H_8(l) + H_2O(g) \rightarrow C_4H_9OH(l)$$

Standard entropy values:

$S^\ominus$ (C_4H_8(l)) = +310 J K^{-1} mol^{-1}

$S^\ominus$ (H_2O(g)) = +189 J K^{-1} mol^{-1}

$S^\ominus$ (C_4H_9OH(l)) = +228 J K^{-1} mol^{-1}

Using the values above, calculate the standard entropy change for the reaction.

Solution steps	Calculations
Step 1: Write out the values given in the	$S^\ominus\ (\mathrm{C_4H_8(l)}) = +310\ \mathrm{J\ K^{-1}\ mol}$

Solution steps	Calculations
question.	$S^{\ominus}\ (\mathrm{H_2O(g)}) = +189\ \mathrm{J\ K^{-1}\ mol}$ $S\ominus\ (\mathrm{C_4H_9OH(l)}) = +228\ \mathrm{J\ K^{-}}$
Step 2: Write out the equation.	$\Delta S^{\ominus} = \Sigma S^{\ominus}\ (\text{products}) - \Sigma S^{\ominus}$
Step 3: Substitute the values given.	$= 228 - (310 + 189)$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$= -271\ \mathrm{J\ K^{-1}\ mol^{-1}}$ Note that the negative sign show: mole of liquid and one mole of ga

Worked example 12

Consider the combustion of propene and the following standard entropy values:

$$C_3H_6(l) + 4.5O_2(g) \rightarrow 3CO_2(g) + 3H_2O(g)$$

Standard entropy values:

$S^{\ominus}\ (C_3H_6(l)) = +267\ \mathrm{J\ K^{-1}\ mol^{-1}}$

$S^{\ominus}\ (O_2(g)) = +205\ \mathrm{J\ K^{-1}\ mol^{-1}}$

$S^{\ominus}\ (H_2O(g)) = +189\ \mathrm{J\ K^{-1}\ mol^{-1}}$

$S^{\ominus}\ (CO_2(g)) = +214\ \mathrm{J\ K^{-1}\ mol^{-1}}$

Using the values above, calculate the standard entropy change for the reaction.

Solution steps	Calculations
Step 1: Write out the values given in the question.	$S^{\ominus}\ (\mathrm{C_3H_6(l)}) = +267\ \mathrm{J\ K^{-1}\ mol^{-}}$ $S^{\ominus}\ (\mathrm{O_2(g)}) = +205\ \mathrm{J\ K^{-1}\ mol^{-1}}$ $S^{\ominus}\ (\mathrm{H_2O(g)}) = +189\ \mathrm{J\ K^{-1}\ mol^{-}}$ $S^{\ominus}\ (\mathrm{CO_2(g)}) = +214\ \mathrm{J\ K^{-1}\ mol^{-}}$

Solution steps	Calculations
Step 2: Write out the equation.	$\Delta S^{\ominus} = \Sigma S^{\ominus}\,(\text{products}) - \Sigma S^{\ominus}$
Step 3: Substitute the values given.	$= 3(+214) + 3(+189) - [(+267)$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding	$= +19.5\ \mathrm{J\ K^{-1}\ mol^{-1}}$ Note that the positive sign shows a mole of liquid and 4.5 moles of ga

Complete the summary table in **Interactive 3** by dragging the listed reactions to the correct space.

Δ*S* negative (entropy decreases)	**Δ*S* positive (entropy increases)**
Changes from gas → liquid → solid	Changes from solid → liquid → gas
Freezing	Melting
Condensation	Vaporisation
Deposition	Sublimation
Crystallisation	Dissolution
Decreasing number of moles of gas	Increasing number of moles of gas

- $H_2O(s) \rightarrow H_2O(l)$
- $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$
- $CO_2(s) \rightarrow CO_2(g)$
- $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
- $H_2O(l) \rightarrow H_2O(g)$
- $NaCl(aq) \rightarrow NaCl(s)$
- $H_2O(g) \rightarrow H_2O(l)$
- $H_2O(l) \rightarrow H_2O(s)$
- $NaCl(s) \rightarrow NaCl(aq)$
- $I_2(g) \rightarrow I_2(s)$

Check

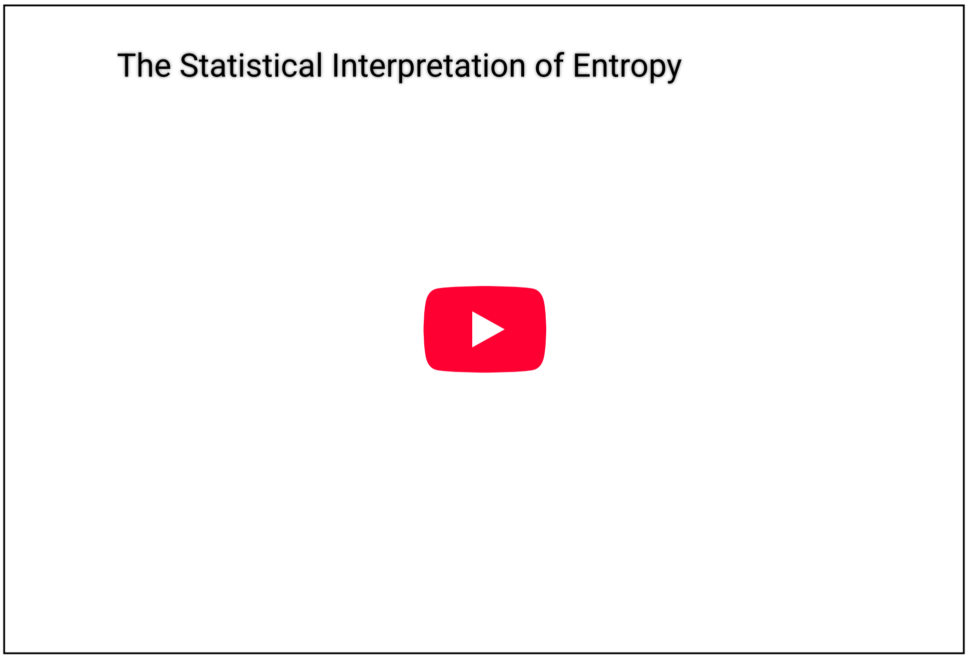
Interactive 3. Complete the Summary Table.

Theory of Knowledge

Entropy is the measurement of disorder, which is constantly increasing in the universe.

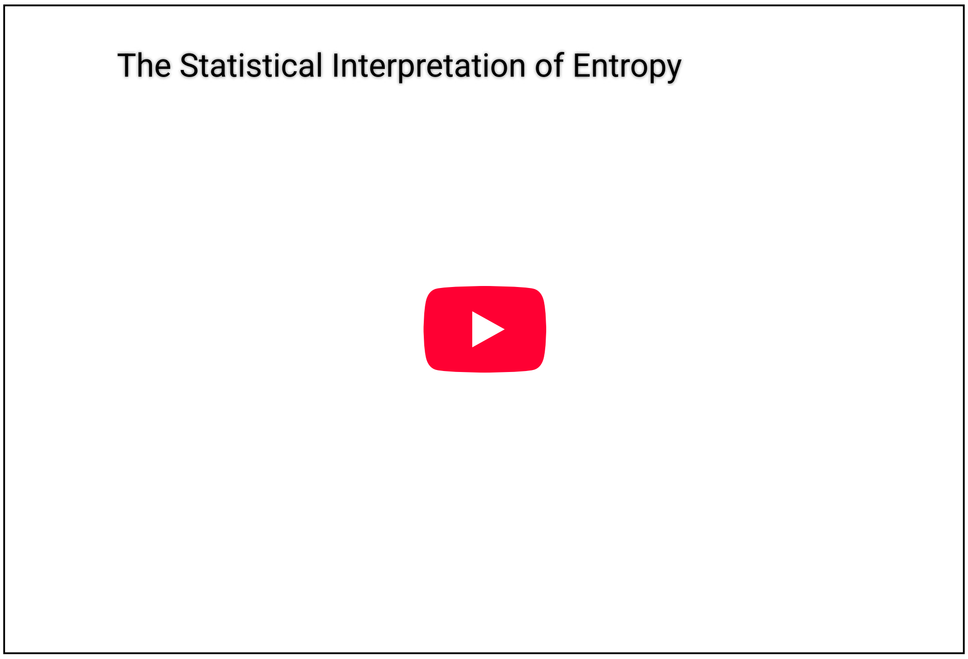
What knowledge is there to be gained by measuring something which is always predictably increasing?

Watch **Video 2** for a visual representation of a model entropy system in a car.



Video 2. Entropy explained through a model car.

Now watch **Video 3** to hear the conclusion that can be made from the model.



Video 3. Conclusions from modelling entropy.

Complete the activity below to practise the content studied in this section.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Communication skills — Using terminology, symbols and communication conventions consistently and correctly
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/whole class activity

Create your own model for entropy:

- Select one example from everyday life where there is a high or low entropy (e.g. a messy room or the ingredients before a meal is prepared).
- Take a photo of your original system and explain why it has high or low entropy.
- Produce a change in entropy in your system (e.g. clean up your room, or prepare a meal with the ingredients selected).
- Take a photo of your new system and explain why its entropy has increased or decreased when compared to your original system.
- Present your work in class.